



RABBIT HOLE · USABILITY EVALUATION

Heuristic & Usability Evaluation Pack

Eight complementary methods · fillable PDF · for the popup and the full dashboard

This pack lets a team **inspect** Rabbit Hole against recognised usability heuristics, **observe** real people using it, **measure** how usable and trustworthy it feels, and **audit** accessibility and failure behaviour — then **consolidate** everything into one prioritised list. Every box on a blue background is a fillable field. It is a **template**: it contains no results, and nothing in it should be read as evidence that the product passes any method.

How the pack fits together

	Method	Who	Time	What you produce	Page
A	Heuristic evaluation (Nielsen's 10)	3–5 evaluators, independently	60–90 min	Severity-rated problems per heuristic	p. 3
B	Cognitive walkthrough	2–3 evaluators	45 min	Step-level learnability failures for 7 tasks	p. 6
C	Moderated think-aloud test	5 target users + facilitator + observer	30–40 min each	Observer log, task success, time, SEQ	p. 9
D	System Usability Scale (SUS)	The same 5 users	3 min each	One 0–100 usability score per user	p. 12
E	Relevance & trust survey	The same 5 users	10 min each	Precision@5, Likert ratings, NPS	p. 13
F	Accessibility audit (WCAG 2.2 AA)	1–2 evaluators + tools	45 min	Pass / fail per success criterion	p. 15
G	Error-state & performance matrix	1–2 evaluators	30 min	Expected vs observed behaviour, timings	p. 17
H	A/B experiment plan	Product team	30 min	Four pre-registered experiments	p. 19
9	Consolidation & sign-off	Facilitator	45 min	Merged findings, matrix, decision	p. 21

Part 10 (the last page) holds the facilitator's hypotheses. **Evaluators must not read it until they have finished Methods A and B.**

Severity scale (use it everywhere)

Rating	Meaning	Think about
0	Not a usability problem	You looked at it and disagree that it is one.
1	Cosmetic	Fix only if there is spare time.
2	Minor	Low priority; slows or mildly annoys a few people.
3	Major	Important to fix; many people stumble, or a few fail the task.
4	Catastrophic	Must be fixed before release; blocks the task or damages trust / privacy.

Rate each problem on three things and then give **one** number: **frequency** (how many users meet it), **impact** (how hard it is to overcome) and **persistence** (a one-off or every time?). Judge the problem, not your preference.

0

Before you start

Ground rules, test environment and evaluator details

Part 0 · everyone

Ground rules

- **Work alone first.** Do not compare notes until everyone has finished; independent evaluation finds more problems and avoids group-think.
- **Go through the interface twice:** once for the overall flow, once screen by screen and element by element.
- **One problem per row.** Say *where* it is, *what* happens, *why* it matters to a user, and *how* you would fix it. Quote the exact on-screen words.
- **Rate honestly.** “0 — no problem found” is a valid, useful answer. Note strengths too: they tell the team what to keep.
- **Privacy.** Do not type real passwords, API keys or other people’s personal data into this form. Observed sessions need the participant’s consent (Method C).
- **Save a copy per evaluator** (for example RabbitHole-Eval-AB.pdf) and send it to the facilitator.

Test environment. Build with `npm run build`, open `chrome://extensions`, turn on Developer mode, **Load unpacked** → choose `dist/`, and pin the icon. Use a fresh Chrome profile with default settings. Test on: a YouTube video, a Reddit thread, a Substack article, a Netflix title page (optional — needs your own subscription; never share credentials or capture DRM video) and one other website. No network or no subscription? Run `npm run dev:ui` and open `http://localhost:5199/?data=docs` — the same interface with built-in sample data.

Checklist

- ☐ Extension loaded from dist/
- ☐ Fresh profile, default settings
- ☐ Page detection ON
- ☐ Normal network
- ☐ Test pages open
- ☐ Popup at 480 × 600

Evaluator and session

Name

Initials / ID

Date

Start / end time

Role

Experience (years)

Chrome version

OS / device

Extension version

1.0.0

Display / zoom

Data used

☐ Live Reddit / Substack

☐ Sample data (dev preview)

☐ Netflix tested

Input

☐ Mouse

☐ Keyboard only

☐ Screen reader

☐ Touch

Methods you completed

☐ A

☐ B

☐ C

☐ D

☐ E

☐ F

☐ G

☐ H

Familiarity (how well do you know these?)

	1 never	2	3	4	5 daily
Reddit					
Substack / newsletters					
YouTube					
Netflix					

A

Heuristic evaluation — Nielsen’s 10 usability heuristics

Inspect the popup and the dashboard; no users needed

3–5 evaluators · 60–90 min

Walk through the interface at least twice. For each heuristic, either tick **No problem found** or record up to two problems (there is room for more in the notes at the end). Severity uses the 0–4 scale on page 1; use - if you have not rated it.

H1

Visibility of system status

☐ No problem found

The design should always keep users informed about what is going on, through appropriate feedback within a reasonable time.

Check in Rabbit Hole

- Skeleton loaders and “Finding related writing...” while a search runs; is anything ever a blank screen?
- The context bar (“Watching — YouTube”) and the stats line (“Found N relevant posts · topics searched”): do they say what Rabbit Hole understood?
- The toolbar badge when a page is detected; whether stale or cached data is labelled (“flagged”) when Reddit is unavailable.

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H2

Match between system and the real world

☐ No problem found

The design should speak the users’ language: words, phrases and concepts familiar to them, in a natural and logical order.

Check in Rabbit Hole

- Section names — Hot Right Now, Rising Fast, Most Discussed, Across Reddit — and the relevance labels Highly relevant / Very relevant / Related.
- Words such as “detection”, “growth”, “bridge”, “Use this page”, “subreddit”: would a first-time user know them?
- Emoji used as section icons (flame, chart, speech bubble, globe): meaningful, or decorative noise?

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H3

User control and freedom

☐ No problem found

Users often choose functions by mistake and need a clearly marked “emergency exit” to leave the unwanted state without extended dialogue.

Check in Rabbit Hole

- Clearing a search (× button, Esc), “Back to trending”, switching tabs mid-load, closing the popup while loading.
- Turning page detection off per platform; “Clear cache” and “Delete everything” (Settings → Your data): are they reversible or confirmed?
- Can the user always tell how to get out of the Reddit-results view back to the original context?

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H4 Consistency and standards
☐ No problem found

Users should not have to wonder whether different words, situations or actions mean the same thing. Follow platform and industry conventions.

Check in Rabbit Hole

- Button labels and behaviour on cards: “Read on Substack”, “Open Reddit”, “Find deeper reading →”, “See what Reddit thinks →”.
- Popup versus full dashboard: same names, same order, same controls? Light versus dark theme.
- Keyboard conventions: “/” focuses search, ← → switch tabs, Esc clears. Do they match what users know from other tools?

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H5 Error prevention
☐ No problem found

Good error messages matter, but the best designs carefully prevent problems from occurring in the first place.

Check in Rabbit Hole

- Nothing is read from an unsupported page until the user presses “Use this page”; optional permissions are requested only when a feature needs them.
- API-key fields (masked, write-only) and the destructive “Delete everything” button: can they be triggered by accident?
- Empty or whitespace-only searches; searching while offline; double-clicking a bridge button.

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H6 Recognition rather than recall
☐ No problem found

Minimise memory load by making elements, actions and options visible. The user should not have to remember information from one part to another.

Check in Rabbit Hole

- Is the current context (what is being searched) always visible while reading results?
- Topic chips, tab counts and labels versus hidden gestures; is the “/” shortcut discoverable (placeholder, tooltip)?
- Does the “Why this is relevant” box show the reason on the card, or must the user remember what the label means?

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H7 Flexibility and efficiency of use
☐ No problem found

Shortcuts — hidden from novices — speed up interaction for experts, so the design serves both inexperienced and experienced users.

Check in Rabbit Hole

- Auto-selecting Related Reading on a video and Trending elsewhere; one-click topic filters; one-click bridges between Reddit and Substack.
- Keyboard-only use (/, Enter, Esc, ← →, Home/End); the full dashboard for wide screens; AI layer and API keys for power users.
- How many steps from “watching a video” to “reading a relevant article”? (Target for task T1: ≤ 30 s.)

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H8 Aesthetic and minimalist design
☐ No problem found

Interfaces should not contain information that is irrelevant or rarely needed. Every extra unit of information competes with the relevant units.

Check in Rabbit Hole

- Information density of a card at 480 × 600: label, source, date, excerpt, tags, “Why”, buttons. What could go?
- Visual hierarchy and contrast in dark and light themes; the colour accent used only for the primary action and status.
- Motion: shimmer, rise-in; is it calm and does it respect reduced-motion settings?

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H9 Help users recognise, diagnose and recover from errors
☐ No problem found

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

Check in Rabbit Hole

- “Reddit blocked the request” state: cause in plain words, one clear next action (Try again · Open Reddit), stale data flagged.
- No results; Netflix labels hidden (“We couldn’t automatically identify what you’re watching”); invalid API key; permission declined.
- Is the user ever blamed, left with a dead end, or shown made-up filler content?

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

H10 Help and documentation
☐ No problem found

It is best if the system needs no extra explanation, but it may be necessary to provide documentation to help users complete tasks.

Check in Rabbit Hole

- “What Rabbit Hole used” on the context bar, the “Private by design” footer and Settings → Privacy: findable and understandable at first use?
- Help text in Settings (especially API & Integrations and Your data); how to obtain a search key; tooltips explaining relevance.
- Is there any first-run explanation of what the extension reads and sends? (Facilitator note: see Part 10.)

Where (screen / element)	Problem — what happens and why it matters	Suggested fix	Severity
			-
			-

Strengths worth keeping

Other problems or overall impression

B

Cognitive walkthrough

Predict where a first-time user will go wrong, step by step

2–3 evaluators · 45 min

Take the role of a **first-time user who knows what they want but has never seen Rabbit Hole** (the curious viewer or the trend watcher from the project report). For every step of every task, answer the four questions below. **Tick the box when the answer is YES.** Every unticked box is a predicted failure: write the “failure story” — who would fail, why, and what they would do instead.

Question	Ask yourself	Look for
Q1	Will the user try to achieve the right effect?	Do they know this step is needed?
Q2	Will the user notice that the correct action is available?	Is the control visible?
Q3	Will the user associate the action with the effect they want?	Does the label / icon make sense?
Q4	After the action, will the user see that progress is being made?	Is there clear feedback?

T1–T5 are the core tasks (they are measured again in Method C); T6 and T7 cover the fallback and failure paths.

T1

From a video to a related article · core task

Scenario.

You are watching a YouTube video about AI agents and want something worth reading about it.

Done when:

A related Substack article opens in a new tab.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	Click the Rabbit Hole toolbar icon (a badge shows that the page was recognised).	☐	☐	☐	☐	
2	The popup opens on Related Reading ; the context bar says “Watching — YouTube” with topics.	☐	☐	☐	☐	
3	Scan the article cards (label · publication · date · excerpt) and choose one.	☐	☐	☐	☐	
4	Click Read on Substack on that card.	☐	☐	☐	☐	

Overall for this task

☐ Learnable at first sight

☐ Learnable with effort

☐ Likely to fail

Unticked Q boxes = predicted failure.

T2

Understand why an article is recommended · core task

Scenario.

You see “Highly relevant” on the top card and wonder whether to trust it.

Done when:

You can explain, in your own words, what the label and the reason are based on.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	Find the relevance label on the card.	☐	☐	☐	☐	
2	Find and read the Why this is relevant box under the excerpt.	☐	☐	☐	☐	
3	Check the stats line (“Found N relevant posts · topics searched”) and the matched tags.	☐	☐	☐	☐	
4	Decide whether to open the article — and say what would make you distrust it.	☐	☐	☐	☐	

Overall for this task

☐ Learnable at first sight

☐ Learnable with effort

☐ Likely to fail

Unticked Q boxes = predicted failure.

T3 Find what is trending on Reddit · core task

Scenario. You want to know what people are discussing on Reddit right now in Technology. **Done when:** A thread you consider worth reading opens on Reddit.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	Switch to the Trending Reddit tab (click it, or press ←/→).	☐	☐	☐	☐	
2	Choose the Technology topic chip.	☐	☐	☐	☐	
3	Compare Hot Right Now, Rising Fast, Most Discussed and Across Reddit; read score, comments and the growth label.	☐	☐	☐	☐	
4	Click Open Reddit on a thread.	☐	☐	☐	☐	

Overall for this task ☐ Learnable at first sight ☐ Learnable with effort ☐ Likely to fail Unticked Q boxes = predicted failure.

T4 Bridge between Reddit and Substack · core task

Scenario. You are reading a Reddit thread and want deeper writing on it — then want to see how Reddit reacted to an article. **Done when:** You reach related articles from a thread, and discussion from an article.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	On a Reddit thread page, open Rabbit Hole; it offers Find deeper reading →.	☐	☐	☐	☐	
2	Click it: Related Reading searches for the thread's topic.	☐	☐	☐	☐	
3	Open a Substack article, reopen the popup and click See what Reddit thinks →.	☐	☐	☐	☐	
4	Return with Back to trending and confirm you know where you are.	☐	☐	☐	☐	

Overall for this task ☐ Learnable at first sight ☐ Learnable with effort ☐ Likely to fail Unticked Q boxes = predicted failure.

T5 Check and control what is read · core task

Scenario. You are not sure what this extension looks at and want to switch it off for one site. **Done when:** You can state what is read, stored and sent, and you switched detection off.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	Open What Rabbit Hole used on the context bar or click Private by design in the footer.	☐	☐	☐	☐	
2	Open Settings (gear) → Privacy and read what is read, stored and sent.	☐	☐	☐	☐	
3	Switch page detection off for one platform; reopen the popup on that platform.	☐	☐	☐	☐	
4	Read "Page detection is off" and find Turn on detection .	☐	☐	☐	☐	

Overall for this task ☐ Learnable at first sight ☐ Learnable with effort ☐ Likely to fail Unticked Q boxes = predicted failure.

T6 Search when nothing is detected · optional

Scenario. You are on Netflix (or any other site) where Rabbit Hole cannot tell what you are watching. **Done when:** You find and open a relevant article by searching for a topic.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	Open the popup; read the fallback message ("We couldn't automatically identify what you're watching").	☐	☐	☐	☐	
2	Type a topic into "Search Reddit or Substack..." — or, on any other site, press Use this page .	☐	☐	☐	☐	
3	Press Enter and watch the loading state ("Finding related writing...").	☐	☐	☐	☐	
4	Open a result.	☐	☐	☐	☐	

Overall for this task ☐ Learnable at first sight ☐ Learnable with effort ☐ Likely to fail

Unticked Q boxes = predicted failure.

T7 Recover from a failure and tidy up · optional

Scenario. Reddit is unreachable (DevTools → Network → Offline). Afterwards you want to remove what Rabbit Hole stored. **Done when:** You understand the message, retry successfully and find the data controls.

#	Correct action (what the interface offers)	Q1	Q2	Q3	Q4	Failure story / notes
1	Open Trending while offline; read the message and what it offers.	☐	☐	☐	☐	
2	Check whether cached results are shown and marked as stale.	☐	☐	☐	☐	
3	Go back online and click Try again .	☐	☐	☐	☐	
4	Open Settings → Your data ; read what Clear cache and Delete everything do.	☐	☐	☐	☐	

Overall for this task ☐ Learnable at first sight ☐ Learnable with effort ☐ Likely to fail

Unticked Q boxes = predicted failure.

Walkthrough summary

Most serious failure

Quick wins



Moderated think-aloud usability test

5 users · 30–40 min each

One facilitator talks, one observer logs. Duplicate these pages for each participant.

Participant

Participant ID		Date / time	
Facilitator		Observer	
Age band	-	Occupation	
Reads newsletters	☐ Never ☐ Monthly ☐ Weekly ☐ Daily		
Uses Reddit	☐ Never ☐ Monthly ☐ Weekly ☐ Daily		
Consent	☐ Informed consent given ☐ Written notes OK ☐ Screen / audio recording OK (optional)		
	☐ Told they may stop at any time ☐ No real personal data entered		

Facilitator script (read, do not improvise)

- “We are testing the extension, not you — nothing you do is wrong. Please **think aloud**: say what you are looking at, what you expect, and anything unclear.”
- Neutral prompts only: “What are you thinking?” · “What did you expect to happen?” · “What would you do next?” Do not explain features, hint at buttons or defend the design.
- Hints in 3 levels: **1** general (“Is there anything on the screen that might help?”) · **2** specific area · **3** show it. Record the level; a level-3 hint counts as a failure. Stop a task after 3 minutes.

Before the tasks

“What do you think a tool called Rabbit Hole does?”	
“What would you expect it to read about you?”	

Task log

Time each task from “go” until the success state in Method B (or until you stop it). **SEQ** (Single Ease Question), asked after each task: “Overall, how easy or difficult was this task?” — 1 very difficult ... 7 very easy.

T1 From a video to a related article · target ≤ 30 s

Start End Time (s) Errors Hint level SEQ

Result ☐ Unassisted ☐ With hint (1–2) ☐ Failed / gave up

Observations, quotes, where they hesitated

T2 Understand why an article is recommended

Start End Time (s) Errors Hint level SEQ

Result ☐ Unassisted ☐ With hint (1–2) ☐ Failed / gave up

Observations, quotes, where they hesitated

T3 Find what is trending on Reddit

Start End Time (s) Errors Hint level SEQ

Result ☐ Unassisted ☐ With hint (1–2) ☐ Failed / gave up

Observations, quotes, where they hesitated

T4 Bridge between Reddit and Substack

Start End Time (s) Errors Hint level SEQ

Result ☐ Unassisted ☐ With hint (1–2) ☐ Failed / gave up

Observations, quotes, where they hesitated

T5 Check and control what is read

Start End Time (s) Errors Hint level SEQ

Result ☐ Unassisted ☐ With hint (1–2) ☐ Failed / gave up

Observations, quotes, where they hesitated

Optional tasks T6 / T7 — record result and notes here if you ran them.

Critical incidents

Anything that slowed, confused, surprised or delighted the participant. Link it to a heuristic (H1–H10) or WCAG if you can.

Time	Task	What happened (and what they said)	Heur.	Sev.
	-		-	-
	-		-	-
	-		-	-
	-		-	-
	-		-	-
	-		-	-
	-		-	-
	-		-	-

Debrief (after the tasks)

What was the most useful part? Why?

What was the most confusing or annoying part?

What do you think Rabbit Hole reads about you, and where does it send it? (compare with Part 10 facts)

Did anything make you trust or distrust the recommendations?

If you could change one thing first, what would it be?

Would you use it again — when, and for what?



System Usability Scale (SUS)

Ten statements, answered straight after the tasks, without discussion

same 5 users · 3 min each

Participant ID

Date

Participant instructions: "Please answer every statement, based on your experience with Rabbit Hole just now. Give your immediate reaction rather than thinking for long."

#	Statement	1 Strongly disagree	2	3	4	5 Strongly agree	Score
1	I think that I would like to use Rabbit Hole frequently.	☐	☐	☐	☐	☐	
2	I found Rabbit Hole unnecessarily complex.	☐	☐	☐	☐	☐	
3	I thought Rabbit Hole was easy to use.	☐	☐	☐	☐	☐	
4	I think that I would need the support of a technical person to be able to use Rabbit Hole.	☐	☐	☐	☐	☐	
5	I found the various functions in Rabbit Hole were well integrated.	☐	☐	☐	☐	☐	
6	I thought there was too much inconsistency in Rabbit Hole.	☐	☐	☐	☐	☐	
7	I would imagine that most people would learn to use Rabbit Hole very quickly.	☐	☐	☐	☐	☐	
8	I found Rabbit Hole very cumbersome to use.	☐	☐	☐	☐	☐	
9	I felt very confident using Rabbit Hole.	☐	☐	☐	☐	☐	
10	I needed to learn a lot of things before I could get going with Rabbit Hole.	☐	☐	☐	☐	☐	

Scoring (facilitator)

- **Odd-numbered** statements (1, 3, 5, 7, 9): contribution = answer – 1. **Even-numbered** (2, 4, 6, 8, 10): contribution = 5 – answer.
- Add the ten contributions (0–40), then multiply by **2.5**: the **SUS score is 0–100**. It is a score, not a percentage.
- Reading it: about **68** is average across products; roughly **80** and above is in the top tenth; below about **51** is poor (Sauro & Lewis, 2016). With five users, report the range as well as the mean.

Sum of contributions

× 2.5 = SUS

Band



< 51



51–67



68–79



≥ 80

All participants

P1	P2	P3	P4	P5	Mean	Min–max



Relevance & trust survey

Do the recommendations deserve trust? Does the person understand what the tool does?

same 5 users · 10 min each

Participant ID

Date

Part 1 — Precision@5

Open two different videos or pages. For each, read the **top five** articles in Related Reading and give **your own judgement** of relevance to the page *before* looking at the relevance label. Record the title exactly as shown; never invent or paraphrase. Target: lenient precision $\geq 70\%$.

A Context A				
Page / video title		Topics in the context bar		
#	Article title — publication (as shown)	Is it relevant to this page?	"Why" box was...	Label vs your view
1		☐ Relevant ☐ Partly ☐ Not	-	-
2		☐ Relevant ☐ Partly ☐ Not	-	-
3		☐ Relevant ☐ Partly ☐ Not	-	-
4		☐ Relevant ☐ Partly ☐ Not	-	-
5		☐ Relevant ☐ Partly ☐ Not	-	-
Relevant Partly Not		Precision@5 lenient = (relevant + partly) / 5 strict = relevant / 5		

B Context B				
Page / video title		Topics in the context bar		
#	Article title — publication (as shown)	Is it relevant to this page?	"Why" box was...	Label vs your view
1		☐ Relevant ☐ Partly ☐ Not	-	-
2		☐ Relevant ☐ Partly ☐ Not	-	-
3		☐ Relevant ☐ Partly ☐ Not	-	-
4		☐ Relevant ☐ Partly ☐ Not	-	-
5		☐ Relevant ☐ Partly ☐ Not	-	-
Relevant Partly Not		Precision@5 lenient = (relevant + partly) / 5 strict = relevant / 5		

Part 2 — Trending quality

#	Thread title (top 5 in "Hot Right Now")	Genuinely trending / worth a look?	Growth label helped?
1		☐ Yes ☐ Unsure ☐ No	☐ Yes ☐ No ☐ No label
2		☐ Yes ☐ Unsure ☐ No	☐ Yes ☐ No ☐ No label
3		☐ Yes ☐ Unsure ☐ No	☐ Yes ☐ No ☐ No label
4		☐ Yes ☐ Unsure ☐ No	☐ Yes ☐ No ☐ No label
5		☐ Yes ☐ Unsure ☐ No	☐ Yes ☐ No ☐ No label

Part 3 — Understanding and trust (Likert, 1–5)

#	Statement	1 Strongly disagree	2	3	4	5 Strongly agree
1	I understood what Rabbit Hole reads from the pages I visit.	☐	☐	☐	☐	☐
2	I understood what Rabbit Hole sends to other services.	☐	☐	☐	☐	☐
3	I felt in control of what Rabbit Hole does (for example, I could switch detection off).	☐	☐	☐	☐	☐
4	The recommended articles were relevant to what I was watching or reading.	☐	☐	☐	☐	☐
5	The relevance labels (Highly / Very relevant / Related) matched my own judgement.	☐	☐	☐	☐	☐
6	“Why this is relevant” helped me decide whether to open an article.	☐	☐	☐	☐	☐
7	The Trending tab showed me something I would not have found myself.	☐	☐	☐	☐	☐
8	Moving between Reddit and Substack (“Find deeper reading”, “See what Reddit thinks”) felt natural.	☐	☐	☐	☐	☐
9	I trust Rabbit Hole not to make things up (titles, authors, dates, links).	☐	☐	☐	☐	☐
10	I would use Rabbit Hole at least weekly.	☐	☐	☐	☐	☐

Mean of items 1–10

Items 1–3 (privacy understanding) mean — target ≥ 4.0

Part 4 — Net Promoter Score

“How likely are you to recommend Rabbit Hole to a friend or colleague?”

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

0 = not at all likely

10 = extremely likely

Detractors 0–6 · passives 7–8 · promoters 9–10. NPS = % promoters – % detractors. With five users treat it as a conversation starter, not a statistic.

Part 5 — In their own words

What did you like most?

What confused or annoyed you?

Did anything feel intrusive, “creepy” or untrustworthy?

What is missing?

Describe Rabbit Hole in three words.

**Accessibility audit — WCAG 2.2 level AA**

Tools plus manual checks; popup and full dashboard, dark and light themes

1–2 evaluators · 45 min

Tools and passes completed

☐ axe DevTools
 ☐ Lighthouse accessibility
 ☐ Keyboard-only pass
 ☐ Screen reader (VoiceOver / NVDA)
 ☐ 200 % zoom
☐ 320 px reflow
 ☐ Contrast tool
 ☐ Reduced-motion on
 ☐ Forced colours

Lighthouse a11y score axe: critical / serious
 Screen reader + browser Themes tested

SC	Criterion — what to check in Rabbit Hole	Pass	Fail	N/A	Evidence / notes
1.1.1	Non-text content. Icon-only buttons (settings gear, dashboard, ×, theme) have accessible names; emoji section icons are labelled or hidden from screen readers.	☐	☐	☐	
1.3.1	Info and relationships. Tabs, chips, lists and headings expose their structure (tablist/tab, headings, list items) — check with the accessibility tree.	☐	☐	☐	
1.3.2	Meaningful sequence. Reading order of the popup and dashboard matches the visual order (screen-reader pass).	☐	☐	☐	
1.4.1	Use of colour. Stale / growth / error / selected states are conveyed by text or shape, not colour alone.	☐	☐	☐	
1.4.3	Contrast (minimum). Text ≥ 4.5:1 (3:1 for large text) in dark, light and system themes — measure with DevTools or axe, including secondary and placeholder text.	☐	☐	☐	
1.4.4	Resize text. Dashboard and Settings at 200 % browser zoom: no loss of content or function.	☐	☐	☐	
1.4.10	Reflow. Dashboard / Settings at 320 CSS px width: no two-dimensional scrolling.	☐	☐	☐	
1.4.11	Non-text contrast. Focus rings, toggles, chip outlines and input borders reach 3:1 against their background.	☐	☐	☐	
1.4.12	Text spacing. Apply a text-spacing override (line 1.5, paragraph 2×, letter 0.12 em, word 0.16 em): nothing clipped or overlapping.	☐	☐	☐	
1.4.13	Content on hover or focus. Tooltips (e.g. relevance explanations) are dismissible, hoverable and persistent.	☐	☐	☐	
2.1.1	Keyboard. Everything works without a mouse: tabs, topic chips, search, card buttons, every Settings control, “Delete everything”.	☐	☐	☐	
2.1.2	No keyboard trap. Focus can always leave the popup and any dialog (Tab, Shift+Tab, Esc).	☐	☐	☐	
2.1.4	Character key shortcuts. “/” focuses search only when no text field has focus, and never fires while typing.	☐	☐	☐	
2.2.2	Pause, stop, hide. No auto-updating or moving content; loading shimmer stops when finished; reduced-motion preference is honoured.	☐	☐	☐	
2.4.3	Focus order. Header → search → tabs → filters → results → footer; no surprising jumps after results load.	☐	☐	☐	
2.4.6	Headings and labels. Headings and control labels describe their purpose (“Open Reddit”, not “Open”).	☐	☐	☐	
2.4.7	Focus visible. A visible focus indicator on every interactive element, in both themes.	☐	☐	☐	

SC	Criterion — what to check in Rabbit Hole	Pass	Fail	N/A	Evidence / notes
2.4.11	Focus not obscured (min). A focused card button is never hidden behind the sticky header or footer (new in WCAG 2.2).	☐	☐	☐	
2.5.8	Target size (minimum). Pointer targets ≥ 24 × 24 CSS px or with enough spacing — measure small icon buttons and chips (new in WCAG 2.2).	☐	☐	☐	
3.1.1	Language of page. Popup, options and dashboard declare their language.	☐	☐	☐	
3.2.2	On input. Changing a filter, toggle or select does not cause an unexpected change of context.	☐	☐	☐	
3.3.1	Error identification. Errors are described in text and say which input or action failed.	☐	☐	☐	
3.3.2	Labels or instructions. Search and key fields have visible labels or instructions; keys are masked but their state is announced.	☐	☐	☐	
4.1.2	Name, role, value. Custom controls expose name, role and state (selected tab, pressed toggle, expanded section).	☐	☐	☐	
4.1.3	Status messages. Result counts, “Finding related writing...” and errors are announced without moving focus (live region).	☐	☐	☐	

Result

Failed criteria

Of which critical

Summary and fixes (target: 0 critical failures)

This checklist is a structured self-audit, not a conformance certification. WCAG 2.2 AA conformance needs a formal audit with representative users.



Error-state & performance matrix

Do failures read well, and is the interface fast enough to feel instant?

1–2 evaluators · 30 min

Part 1 — Error states

Trigger each state, compare with the expected behaviour, and tick what holds. **Cause** = the cause is stated in plain words · **Next** = a next step is offered · **Honest** = nothing is invented or guessed. Result: **Pass**, **Fail** or **Skipped** (say why).

Scenario and how to trigger	Expected behaviour	Cause	Next	Honest	Result	Observed
G1 · Reddit unreachable DevTools → Network → Offline (or block reddit.com), then open Trending.	Plain-language cause (“Reddit blocked / could not be reached”), one clear next step (Try again · Open Reddit); cached data, if any, shown and flagged as stale.	☐	☐	☐	-	
G2 · Topic with no threads Pick the narrowest topic chip with few results.	Honest empty state that says why and what to try; no filler threads.	☐	☐	☐	-	
G3 · No search results Search for a nonsense string such as “zzxqjw”.	Clear “nothing found” message with suggestions; no invented articles; the search can be edited or cleared.	☐	☐	☐	-	
G4 · Page with no usable metadata On a near-empty page press “Use this page”.	Explains that nothing could be identified and offers manual search.	☐	☐	☐	-	
G5 · Netflix labels hidden Open a title page, keep controls hidden (idle mouse) and open the popup.	“We couldn’t automatically identify what you’re watching” + manual search box; never a wrong guess.	☐	☐	☐	-	
G6 · Detection switched off Settings → Privacy → turn detection off for a platform; open that platform.	“Page detection is off” with Turn on detection; nothing is read from the page.	☐	☐	☐	-	
G7 · Optional permission declined Enable “Include custom-domain newsletters” and deny the browser prompt.	The feature stays off, nothing breaks, and the explanation is understandable.	☐	☐	☐	-	
G8 · Invalid search key Settings → API & Integrations: enter an obviously invalid search key.	Search falls back to curated feeds; the key is never shown back; the message is understandable.	☐	☐	☐	-	
G9 · Invalid AI key (if used) Enable the AI layer with an invalid key.	Falls back to template explanations; no crash; no key echoed.	☐	☐	☐	-	
G10 · Slow network DevTools → Network → Slow 3G; open Related Reading.	Skeletons, never a blank screen; the popup stays responsive; closing it cancels requests.	☐	☐	☐	-	
G11 · Cold service worker chrome://extensions → inspect the service worker → stop it, then open the popup.	The popup works on first try; no stuck loader.	☐	☐	☐	-	

Part 2 — Performance (proposed targets; calibrate on your machine)

Measure with DevTools → Performance (inspect the popup) or a 60 fps screen recording, counting frames. Take three runs, quit nothing in between, and record the median. Reddit and Substack speed is outside the product’s control: judge P3 and P4 against the network you used.

Metric	Target	Run 1	Run 2	Run 3	Median	Meets?
P1 Popup first paint (skeleton visible) after clicking the icon	≤ 300 ms					-
P2 Trending threads visible, cache warm	≤ 500 ms					-
P3 Trending threads visible, cold network	≤ 3 s					-
P4 Related Reading: first cards visible, cold	≤ 6 s					-
P5 Related Reading: cards visible, cache warm	≤ 500 ms					-
P6 Typing a new query while one runs: the old request is cancelled and no stale results flash	pass / fail					-
P7 Popup closed and idle for 60 s: no network or CPU activity apart from the optional background refresh	quiet					-

Measurement conditions

Machine / CPU

Network

Chrome version

Other extensions on

Observations, anomalies, screenshots taken

Summary: biggest risk found in Method G

H

A/B experiment plan

Four design questions, written down before anyone looks at results

product team · 30 min

No telemetry — by design. Rabbit Hole collects no analytics, so experiments cannot be measured from usage data. Run them as **moderated** or **unmoderated task studies** (with the metrics below captured by an observer, a screen recording or a survey), or as two builds given to different participants. Pre-register the metric and the decision rule; about 20 participants per variant lets you spot large differences, not small ones.

The four experiments are pre-filled from the design hypotheses; edit anything. Replace *Decision rule* with concrete numbers before starting.

E1

Show the basis of a relevance label · tests H-1, H-3, H-7

Hypothesis	Showing the matched topics on the card, instead of only inside a hidden tooltip, raises understanding and click-through without slowing the first click.		
Variant A	Control: label (Highly / Very relevant / Related) with the explanation in a tooltip.		
Variant B	Variant: same label plus a one-line, always-visible basis ("Matches: AI agents, software development").		
Primary metric	Share of participants who open an article; Likert "I understand why this was recommended" (1-5).		
Guardrails	Time to first click; error / abandon rate.		
Design	Participants / variant	>= 20	Duration / sessions Run as Moderated Unmoderated Two builds

E2

First-run explanation of what is read · tests H-2

Hypothesis	A one-time, dismissible note about what the extension reads and sends raises privacy understanding without hurting task success.		
Variant A	Control: no first-run message (privacy found via the footer and Settings).		
Variant B	Variant: one-time note on first open, "Rabbit Hole reads only the page you have open, when you open it. What it reads", dismissible, never shown again.		
Primary metric	Likert "I understand what Rabbit Hole reads and sends" (target mean >= 4.0).		
Guardrails	Dismiss time; T1 completion; whether people read it.		
Design	Participants / variant	>= 20	Duration / sessions Run as Moderated Unmoderated Two builds

E3 Which tab opens first on a video page · tests H-6			
Hypothesis	Opening on Related Reading when a video is detected is a better default than opening on Trending, because users came to find related reading.		
Variant A	Control: Related Reading first when a video is detected.		
Variant B	Variant: Trending first, with the "Because you are watching" rows pinned at the top.		
Primary metric	First-click success on T1 and T3; time to first meaningful click.		
Guardrails	Tab switches before first click; stated surprise.		
Design	Participants / variant	≥ 20	Duration / sessions Run as ☐ Moderated ☐ Unmoderated ☐ Two builds
Decision rule	Decide before running. Example: ship B if its primary metric beats A by the agreed margin and no guardrail worsens.		
Results	A	B	Spread / CI
Decision	☐ Ship B ☐ Keep A ☐ Iterate ☐ Inconclusive		

E4 Make the Netflix fallback the obvious next step · tests H-4			
Hypothesis	Putting the search box in focus with a short prompt makes people use the fallback sooner than the current message does.		
Variant A	Control: message plus manual search box below it.		
Variant B	Variant: message plus focused search box with the prompt "Type the title you are watching" and a primary Search button.		
Primary metric	Share who start a manual search within 10 s; time to first search.		
Guardrails	Abandon rate; wrong-title searches.		
Design	Participants / variant	≥ 20	Duration / sessions Run as ☐ Moderated ☐ Unmoderated ☐ Two builds
Decision rule	Decide before running. Example: ship B if its primary metric beats A by the agreed margin and no guardrail worsens.		
Results	A	B	Spread / CI
Decision	☐ Ship B ☐ Keep A ☐ Iterate ☐ Inconclusive		

9

Consolidation & sign-off

Merge every evaluator's findings into one list, then decide

facilitator · 45 min

9.1 Merged findings log

One row per distinct problem. Merge duplicates, quote the on-screen words, and take severity as the median of the evaluators' ratings (discuss any gap of two or more points). Priority: **P1** = severity ≥ 3, or found by two or more methods; **P2** = severity 2; **P3** = severity 0–1.

ID	Source	Problem (observation → consequence)	Methods	Sev.	Prio.	Fix / owner / status
1	-			-	-	
2	-			-	-	
3	-			-	-	
4	-			-	-	
5	-			-	-	
6	-			-	-	
7	-			-	-	
8	-			-	-	
9	-			-	-	
10	-			-	-	
11	-			-	-	
12	-			-	-	
13	-			-	-	
14	-			-	-	
15	-			-	-	
16	-			-	-	
17	-			-	-	
18	-			-	-	

9.2 Heuristic × evaluator severity matrix (Method A)

Enter each evaluator’s **worst** severity for the heuristic (0–4). “Max” shows where the greatest risk lies; “Agree” counts evaluators who rated it ≥ 2.

Heuristic	E1	E2	E3	E4	E5	Max	Agree	# problems
H1 Visibility of system status	-	-	-	-	-			
H2 Match between system and the real world	-	-	-	-	-			
H3 User control and freedom	-	-	-	-	-			
H4 Consistency and standards	-	-	-	-	-			
H5 Error prevention	-	-	-	-	-			
H6 Recognition rather than recall	-	-	-	-	-			
H7 Flexibility and efficiency of use	-	-	-	-	-			
H8 Aesthetic and minimalist design	-	-	-	-	-			
H9 Help users recognise, diagnose and recover from errors	-	-	-	-	-			
H10 Help and documentation	-	-	-	-	-			

Evaluators

Overall mean severity

9.3 User-test results (Methods C, D, E)

Result key: **U** unassisted · **H** with hint · **F** failed. Report counts (“4 of 5”), not percentages, with five users.

	T1	T2	T3	T4	T5	T1 time (s)	SEQ mean	SUS	P@5	NPS 0–10
P1	-	-	-	-	-					
P2	-	-	-	-	-					
P3	-	-	-	-	-					
P4	-	-	-	-	-					
P5	-	-	-	-	-					
Summary										

9.4 Against the success criteria

Metric	Target	Measured	Met?
Task completion, core tasks T1-T5 (Method C)	>= 80 % unassisted		☐ Yes ☐ No ☐ n/a
T1 time: open on a video to open a related article (C)	<= 30 s median		☐ Yes ☐ No ☐ n/a
SUS (D)	>= 75		☐ Yes ☐ No ☐ n/a
Precision@5, lenient (E)	>= 70 %		☐ Yes ☐ No ☐ n/a
Heuristic evaluation (A): major (3) or catastrophic (4) problems	0 unresolved before release		☐ Yes ☐ No ☐ n/a
Accessibility audit (F): critical failures	0		☐ Yes ☐ No ☐ n/a
"I understand what Rabbit Hole reads and sends" (E, items 1-3)	>= 4.0 mean		☐ Yes ☐ No ☐ n/a

9.5 Top findings

Problem 1

Problem 2

Problem 3

Problem 4

Problem 5

Keep 1

Keep 2

Keep 3

9.6 Decision

☐ Ship as is ☐ Ship after all P1 items are fixed ☐ Hold the release ☐ Fix, then re-test

Rationale and next steps

Sign-off

Name	Role	Date	Initials / signature

10

Facilitator notes

Hypotheses, analysis guidance and references

do not read before you have finished A and B

Why this page is last. The design team goes in with eight suspicions. If evaluators see them first, they will look only where the team points and agree with it. Reveal this page to evaluators after they have completed Methods A and B independently; a hypothesis that nobody found unprompted is itself a finding.

ID	Hypothesis	Where to look	What would confirm it	Methods
H-1	"Highly relevant" is not explained — the tooltip that says how it was judged is hidden.	T2; the label and the "Why" box on a card.	≥ 2 of 5 users cannot say what the label is based on; evaluators flag heuristics 6 and 10.	A, B, C, E
H-2	There is no first-run explanation of what the extension reads; privacy clarity depends on finding the footer or Settings.	First open; T5; Likert items 1–3.	Mean < 4.0 on items 1–3, or users need > 60 s to find Privacy.	A, C, E
H-3	Cards are dense at 480 × 600 and the "Why this is relevant" box is skipped.	Card layout; observation during T1–T2.	≥ 3 of 5 never mention the "Why" box unprompted.	A, C
H-4	Netflix users whose labels are hidden do not notice the fallback search box.	T6 on a Netflix title with hidden controls; G5.	Users wander or click elsewhere for > 10 s before using the search box.	B, C, G
H-5	Emoji section icons (flame, chart, speech bubble, globe) are ambiguous or noisy for screen-reader users.	Topic / section chips with a screen reader; SC 1.1.1, 1.3.1.	The screen reader reads emoji names, or the chips' purpose is unclear.	F
H-6	The automatic tab choice (Related Reading on a video) surprises people who expect Reddit first.	The first popup open on a video (T1) and then T3.	≥ 2 users say they expected Reddit first, or switch tabs immediately.	B, C
H-7	"Because you're watching" rows can show weak matches, hurting trust in the whole tab.	Trending tab while a video is detected; Method E part 2.	More than 1 in 5 of those rows are judged not relevant.	E
H-8	The Settings page (six sections) is long; users may not find "Delete everything" or the key fields.	T5 and T7; Settings → Your data.	More than 45 s, or visible hunting, to find the control.	B, C

Facts for checking participants' understanding (Method C debrief, Method E items 1–2)

- **Read:** only the page that is open when the user opens the popup, and only on YouTube, Netflix, Reddit and Substack — or on any other site when the user presses "Use this page".
- **Stored:** settings and any API keys on the device; caches in memory only, cleared when the browser closes; no browsing history.
- **Sent:** short topic queries to Reddit and a search provider; with the optional AI layer, a title, channel and a short description excerpt. No analytics, no tracking.

Analysing the results

- Individual evaluators find only a minority of problems; the union of three to five finds many more — that is why Method A is run independently and merged (Nielsen & Landauer, 1993).
- Write each problem as **observation** → **consequence** → **heuristic**. Quote the interface. Separate the problem from the proposed fix.
- Five users are enough to surface most *qualitative* problems, but not to estimate rates. Report counts, ranges and quotes, not percentages or significance.
- Triangulate: a problem seen in A and B and C is far more credible than one seen once. Mark such rows P1 in the findings log.
- Typical benchmarks: SUS ≈ 68 average; SEQ ≈ 5.5 average; NPS depends heavily on context — compare with your own earlier rounds rather than with the industry.
- **Limits of this pack:** inspection methods cannot replace users; the form is a template and the extension has not yet been evaluated with it.

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